

Counting Colored Linear Snake Polyominoes Up to Symmetry

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Abstract

I count colored snake polyominoes in their linear arrangement: L cells in a straight line, each with binary edge colors and degree constraint $\sum \leq 2$. The single linear shape isolates the coloring problem. A 2-state transfer matrix yields raw counts; a Burnside-style average over the line stabilizer (with global flip) gives orbit counts. The worked example $L = 2$ produces 65 raw colorings and 21 distinct orbits. The linear orbit sequence starts 4, 21, 109, 586, 3326, $\dots$ and matches brute force and project tests through $L = 15$.

1 Linear model

Shape: L consecutive unit squares on a line. Each cell has edge bits $(S, E, N, W) \in \{0, 1\}^4$ with degree constraint $\sum \leq 2$, giving 11 admissible patterns. Adjacent cells must agree on the shared hinge edge. With geometry fixed, the task reduces to counting colorings and collapsing symmetries.

2 Symmetry

For $L > 1$ the geometric stabilizer G_0 has 4 elements (identity, path reversal, reflection across the line, reflection composed with reversal); for $L = 1$, $|G_0| = 8$. Adding global color flip yields acting group $G = G_0 \times \{\text{id}, \text{flip}\}$. Distinct linear snakes are orbits of colorings under G .

3 Transfer matrix

Only the hinge bit (0 or 1) passes between cells. Over the 11 admissible patterns, the West→East transfer matrix is

$$M = \begin{pmatrix} 4 & 3 \\ 3 & 1 \end{pmatrix},$$

rows = entry bit, columns = exit bit. Endpoint vectors are $v_{\text{in}} = v_{\text{out}} = (7, 4)$. Raw colorings:

$$\text{Raw}(L) = v_{\text{in}} M^{L-2} v_{\text{out}}^\top, \quad \text{Raw}(1) = 11.$$

4 Burnside over the line stabilizer

Let $\text{Fix}_V(g)$ be colorings fixed by g with all 11 patterns; $\text{Fix}_{\text{both}}(g)$ restricts to degree-2 patterns (needed for the flip terms). The orbit count is

$$N(L) = \frac{2 \sum_{g \in G_0} |\text{Fix}_V(g)| - \sum_{g \in G_0} |\text{Fix}_{\text{both}}(g)| + \sum_{g \in G_0} |\text{Fix}_{\text{both}}(f \circ g)|}{2 |G_0|},$$

with f the global flip. For $L > 1$, $|G_0| = 4$; for $L = 1$, $|G_0| = 8$.

5 Worked example: $L = 2$

- Raw: $\text{Raw}(2) = 65$.
- $|G_0| = 4$, full group size 8.
- Burnside numerator yields $N(2) = 21$.

So 21 distinct linear colored snakes of length 2.

6 Linear sequence

Orbit counts (transfer + Burnside) for the linear arrangement:

$$\begin{aligned} &4, 21, 109, 586, 3326, 19209, 111871, 653758, \\ &3824678, 22387074, 131052313, 767211817, \\ &4491420695, 26293679325, 153927402355 \quad (L = 1..15). \end{aligned}$$

Raw counts (transfer only) start 11, 65, 380, 2196, 12804, $\dots$; orbit counts are smaller after symmetry collapse.

7 Validation and cost

Tests cover: (i) brute-force orbit vs Burnside for $L \leq 3$; (ii) raw vs transfer for $L \leq 3$; (iii) Burnside numerator divisibility for $L \leq 8$. All pass. Computation for linear counts is dominated by fast powers of the 2×2 matrix; large L is cheap.

8 Beyond the line (context only)

For bent snakes, shapes are enumerated once with pruning; the same coloring + Burnside routine applies per shape, then totals are summed. Those results are secondary here; focus remains the linear sequence above.

9 Outlook

Next steps: degree-exact (sum=2) variants, weighted edges, and extracting characteristic polynomials to study linear-snake asymptotics.

10 Exact recurrence and closed form (linear arrangement)

Because the linear arrangement is counted by a finite-state transfer-matrix/Burnside computation, the orbit sequence $N(L)$ is *(eventually) linear-recursive*: there exist constants $c_0, \dots, c_{d-1}$ such that

$$N(L) = \sum_{j=0}^{d-1} c_j N(L-1-j) \quad (L \text{ large enough}).$$

In my project, the smallest recurrence I verified for the full linear-orbit sequence has order $d = 12$ and holds for all $L \geq 14$:

$$\begin{aligned} N(L) = & 12N(L-1) - 38N(L-2) - 38N(L-3) + 370N(L-4) - 394N(L-5) \\ & - 556N(L-6) + 1160N(L-7) - 690N(L-8) + 370N(L-9) + 330N(L-10) \\ & - 750N(L-11) + 225N(L-12) \quad (L \geq 14), \end{aligned}$$

with initial values $N(1), \dots, N(13)$ taken from the transfer/Burnside computation (so the recurrence produces exact integers thereafter).

What “closed form” means here. For a constant-coefficient recurrence, the standard closed form is a sum of exponentials

$$N(L) = \sum_{i=1}^m A_i \lambda_i^L,$$

where λ_i are the distinct roots of the characteristic polynomial of the recurrence and the coefficients A_i are determined uniquely by matching enough initial values. In my case the characteristic polynomial factors as

$$(x-3)(x-1)(x^2-3)(x^2-5x-5)(x^2-3x+1)(x^4-5x^2-5),$$

so all λ_i are explicit algebraic numbers (combinations of $\sqrt{3}$ and $\sqrt{5}$, plus a complex-conjugate pair coming from the negative root of x^2-5x-5).

Why the λ^L formula is not exact. The number

$$\lambda = \frac{5+3\sqrt{5}}{2} \approx 5.854101966$$

is the *dominant* characteristic root, so it governs the *asymptotic* growth

$$N(L) \sim C \lambda^L$$

for an explicit constant C . This approximation keeps only the largest-magnitude exponential term and drops the other 11 terms, so it cannot agree *exactly* with integer values for every L . To get an exact closed form one must keep *all* characteristic-root contributions (or equivalently use the exact recurrence with enough initial values).

11 Connection to the six-vertex model

The SIAM News article on active hydraulics describes an experiment where microscopic rollers produce steady currents on a square grid of pipes, and the allowed junction flows match the six-vertex (square ice) rules [2, 3]. The six-vertex model is a prototypical example where transfer matrices are the natural language: local vertex constraints are compiled into a matrix that propagates boundary data across a strip, and global counts or partition functions are traces/powers of that matrix [4, 5, 1].

My linear-snake counting problem is a 1D cousin of this idea. Each square cell is a local “vertex gadget” with a finite list of allowed edge patterns, and the hinge constraint is exactly a local compatibility rule between neighboring gadgets. The 2×2 matrix M is therefore the simplest transfer matrix one can write down: it propagates a single-bit boundary condition (the hinge color) along the line. After that local counting step, Burnside’s lemma plays the role of a symmetry quotient to identify configurations that are the same after reversing the line, reflecting it, or globally flipping colors.

12 Real-world use case (why the linear arrangement matters)

If a physical system is built from identical modules in a line (microfluidic cells, repeated junction blocks, or any 1D chain of locally constrained gadgets), then the central engineering question is often: *how large is the space of realizable global states?* The linear arrangement isolates that question because the geometry is fixed and the only growth comes from combining local admissible patterns subject to hinge-matching. The transfer matrix gives an exact scaling law, and the Burnside quotient reports the number of *physically nonredundant* configurations when the device can be flipped end-to-end or when a global $0 \leftrightarrow 1$ relabeling is an equivalence.

References

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